

Exact Finite Dimer Random Sequential Adsorption on Paths and Cycles

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Abstract

We give a finite-size solution of dimer random sequential adsorption on a path and a simple cycle, viewed as a uniformly random greedy edge order. Arbitrary differentiation produces a triangular system for all factorial moments. Explicit first and second solutions give $\text{Var}(M_n) = e^{-4}n + 2e^{-4} + o(1)$. Binary gap words give the exact path and cycle supports, with every allowed cardinality attained.

1 Model, ownership, and claim boundary

Let P_n have vertices $1, \dots, n$ and edges $e_i = \{i, i+1\}$, with P_0 empty. Assign iid continuous priorities to the edges and inspect them in increasing order. An edge is accepted exactly when both endpoints are currently unmatched. Equivalently, scan a uniformly random permutation of the labeled edges. Let M_n be the number accepted and write

$$F_n(z) = \mathbb{E}z^{M_n}.$$

For the simple labeled cycle C_n , $n \geq 3$, write the corresponding count and PGF as K_n and $G_n(z)$.

The terminal edge set is a jammed *maximal* matching: no further edge can be added. It need not be a *maximum* matching. This distinction is part of every statement below. Continuous priorities merely generate a uniform order; a discrete-priority model needs a separate tie-breaking convention.

One-dimensional blocking goes back at least to Flory's polymer calculation [1]; Evans reviews random and cooperative sequential adsorption [2]. Penrose gives rigorous general jamming-limit context [3], while Dyer and Frieze place randomized greedy matching in its algorithmic lineage [4]. We reconstruct an exact finite theorem and integrity chain. We do not claim invention or literature priority for the model, the limiting density, or random greedy matching.

2 The finite theorem

Theorem 1 (Path/cycle PGFs, moments, support, and boundary effect). *Set $F_0 = F_1 = 1$. For every $n \geq 2$,*

$$(n-1)F_n(z) = z \sum_{a+b=n-2} F_a(z)F_b(z). \quad (1)$$

Thus the formal OGF $F(x, z) = \sum_{n \geq 0} F_n(z)x^n$ is the unique formal solution with $[x^0]F = [x^1]F = 1$ of

$$F_x = \frac{F-1}{x} + zx F^2, \quad xF_x - F + 1 = zx^2 F^2. \quad (2)$$

Put $H_r(x) = \partial_z^r F(x, z)|_{z=1}$. Then $H_0 = (1-x)^{-1}$, and for every $r \geq 1$, $H_r(0) = 0$ and

$$\begin{aligned} H'_r - \left(\frac{1}{x} + \frac{2x}{1-x} \right) H_r &= x \sum_{j=1}^{r-1} \binom{r}{j} H_j H_{r-j} \\ &\quad + rx \sum_{j=0}^{r-1} \binom{r-1}{j} H_j H_{r-1-j}. \end{aligned} \quad (3)$$

This triangular system determines every factorial moment. In particular, for $n \geq 2$,

$$\mathbb{E}M_n = \sum_{j=1}^{n-1} (n-j) \frac{(-1)^{j+1} 2^{j-1}}{j!}. \quad (4)$$

If $a = e^{-2}$ and $\alpha = (1-a)/2$, then

$$\mathbb{E}M_n = \alpha n - a + o(1), \quad \text{Var}(M_n) = a^2 n + 2a^2 + o(1). \quad (5)$$

The support is $\{0\}$ for $n = 0, 1$. For $n \geq 2$,

$$\text{supp}(M_n) = \left\{ \left\lceil \frac{n-1}{3} \right\rceil, \dots, \left\lfloor \frac{n}{2} \right\rfloor \right\}. \quad (6)$$

For every simple cycle $n \geq 3$,

$$G_n(z) = zF_{n-2}(z), \quad \text{supp}(K_n) = \left\{ \left\lceil \frac{n}{3} \right\rceil, \dots, \left\lfloor \frac{n}{2} \right\rfloor \right\}. \quad (7)$$

Consequently

$$\mathbb{E}K_n = 1 + \mathbb{E}M_{n-2} = \alpha n + o(1), \quad (8)$$

$$\mathbb{E}K_n - \mathbb{E}M_n = e^{-2} + o(1), \quad (9)$$

$$\text{Var}(K_n) = \text{Var}(M_{n-2}) = e^{-4}n + o(1). \quad (10)$$

Both expected occupied-vertex fractions tend to $1 - e^{-2}$.

3 First-edge decomposition and the Riccati OGF

The first edge in any order is accepted. Conditional on e_i being first, its endpoints $i, i+1$ are blocked and the eligible residual vertices split into P_{i-1} and P_{n-i-1} . Given this event, the relative orders induced on those two disjoint edge sets are independent and uniform. Their interleaving has no effect because the vertex sets are disjoint. Therefore

$$M_n \stackrel{d}{=} 1 + M_{i-1}^{(L)} + M_{n-i-1}^{(R)} \quad \text{given } e_i \text{ first,}$$

where the two variables on the right are independent. Averaging over the $n-1$ equally likely choices proves (1). Multiplication by x^{n-1} and summation over $n \geq 2$ gives

$$F_x - \frac{F-1}{x} = zxF^2,$$

which is (2). This is a formal-power-series derivation, so it does not presume analytic convergence in two variables.

At $z = 1$, normalization gives $F_n(1) = 1$, hence $F(x, 1) = (1-x)^{-1}$. Differentiate zF^2 exactly r times:

$$\partial_z^r (zF^2)|_{z=1} = \sum_{j=0}^r \binom{r}{j} H_j H_{r-j} + r \sum_{j=0}^{r-1} \binom{r-1}{j} H_j H_{r-1-j}.$$

The $j = 0, r$ terms in the first sum are $2H_0H_r$. Moving them to the left proves the all- r equation (3); all other indices are strictly below r .

For $r = 1$, the equation is

$$H_1' - \left(\frac{1}{x} + \frac{2x}{1-x} \right) H_1 = \frac{x}{(1-x)^2},$$

and its formal solution regular at zero is

$$H_1(x) = \frac{x(1 - e^{-2x})}{2(1-x)^2}. \quad (11)$$

Expanding the exponential and extracting $[x^n]$ proves (4).

4 Second factorial moment and variance

For $r = 2$, the triangular equation becomes

$$H_2' - \left(\frac{1}{x} + \frac{2x}{1-x} \right) H_2 = 4xH_0H_1 + 2xH_1^2. \quad (12)$$

The unique formal solution regular at zero is

$$H_2(x) = \frac{xe^{-4x}(4x^2e^{2x} - 3xe^{4x} - x + e^{4x} - 1)}{4(x-1)^3}. \quad (13)$$

Direct differentiation verifies (12). The formula also follows by multiplying by the integrating factor $e^{2x}(1-x)^2/x$ and integrating from zero.

To extract the variance without fitting finite data, set $s = 1 - x$, $a = e^{-2}$, and $\alpha = (1 - a)/2$. Equations (11) and (13) have pole parts

$$H_1 = \frac{\alpha}{s^2} - \frac{1+a}{2s} + R_1(x), \quad (14)$$

$$H_2 = \frac{c_3}{s^3} + \frac{c_2}{s^2} + \frac{c_1}{s} + R_2(x), \quad (15)$$

where

$$c_3 = \frac{(1-a)^2}{2}, \quad c_2 = -\frac{5}{4} + a + \frac{5}{4}a^2, \quad c_1 = \frac{3}{4} + a + \frac{5}{4}a^2. \quad (16)$$

After these pole parts are removed, R_1, R_2 are entire combinations of exponentials and polynomials divided by removable powers of $1 - x$. Their Taylor coefficients are therefore $o(1)$ (indeed factorially small up to a polynomial factor). It follows that

$$\begin{aligned} \mathbb{E}M_n &= \alpha(n+1) - \frac{1+a}{2} + o(1) = \alpha n - a + o(1), \\ \mathbb{E}(M_n)_2 &= c_3 \binom{n+2}{2} + c_2(n+1) + c_1 + o(1). \end{aligned}$$

Since $c_3/2 = \alpha^2$, substituting these expressions into $\text{Var}(M_n) = \mathbb{E}(M_n)_2 + \mathbb{E}M_n - (\mathbb{E}M_n)^2$ cancels the quadratic term. The linear coefficient simplifies to a^2 and the constant to $2a^2$, proving (5). Thus the requested leading law is $\text{Var}(M_n) = e^{-4}n + O(1)$, with its boundary constant also identified.

5 Exact support: binary gap words

Lemma 1 (Path support). *A cardinality k occurs as a maximal matching of P_n exactly when $2k \leq n \leq 3k + 1$, with the separate empty cases $n = 0, 1$.*

Proof. List the k selected dimers from left to right. There are $k + 1$ gaps of unmatched vertices: two end gaps and $k - 1$ internal gaps. Maximality is equivalent to every gap having size at most one, since two adjacent unmatched vertices would support another edge. Hence

$$n = 2k + g_0 + \cdots + g_k, \quad g_j \in \{0, 1\},$$

which is possible exactly for $0 \leq n - 2k \leq k + 1$. Conversely, choosing any $n - 2k$ of these gaps to have size one explicitly constructs a maximal matching. This proves the equivalence. $\square$

Every such maximal matching occurs with positive probability under RSA: place all of its mutually disjoint edges first in the scan. They are accepted, and every remaining edge meets one of them by maximality, so it is rejected. Lemma 1 therefore proves (6), not merely a bound on possible sizes.

For a cycle with k selected dimers there are k cyclic gaps, each again zero or one. Thus $n = 2k + g_1 + \cdots + g_k$, so $2k \leq n \leq 3k$, and the same ordering argument realizes every constructed matching. This proves the support in (7). Notice that smaller support values are maximal but not maximum; for example, the middle edge alone is a maximal matching of P_4 , while a maximum matching has size two.

6 Cycle identity and the boundary correction

The first edge of C_n is accepted. Removing its two endpoints from further competition leaves exactly a path on $n - 2$ vertices; conditional on the first edge, the residual relative order is uniform. The identity $K_n \stackrel{d}{=} 1 + M_{n-2}$ follows for every $n \geq 3$, including C_3 and C_4 , and proves (7). Taking moments gives (8) and $\text{Var}(K_n) = \text{Var}(M_{n-2})$. Finally,

$$1 + \alpha(n - 2) - a = \alpha n, \quad 1 - 2\alpha - a = 0,$$

which explains the exact cancellation of the path mean’s constant boundary term. Comparison with the same-size path leaves the positive correction $a = e^{-2}$ in (9). The cycle variance has no constant term at this precision because $a^2(n - 2) + 2a^2 = a^2n$.

7 Boundary atlas and theorem limits

Face	Exact treatment	Not silently identified with
P_0, P_1	$M_n = 0, F_n = 1$	the $n \geq 2$ recurrence
P_2	$M_2 = 1, F_2 = z$	a limiting-density statement
C_n	simple cycle, $n \geq 3$	loops or parallel-edge cycles
priority ties	null under continuity	unspecified discrete tie rules
terminal state	maximal matching	guaranteed maximum matching
finite tables	regression evidence	proof for arbitrary n

The theorem concerns the final count. It does not classify adsorption times under an additional physical-time embedding, general graphs, weighted edge orders, correlated priorities, or discrete ties. Such changes destroy at least one symmetry used in the first-edge convolution.

References

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